36118 Applied NLP · Assessment 2A · Spring 2026

Syllabus to Seek

Skillabus closes the gap between your degree and the job ad

We put subject outlines, job ads and resumes into one skills vocabulary, so a student sees which skills a role needs that they lack, and which subject fills each gap.

Quan Pham · Van Thong · Phuong Nhi
Hoai Thuong · Akash

Master of Data Science and Innovation
University of Technology Sydney

1 The problem

Subject outlines speak academic. Job ads speak industry. Nothing maps one to the other.

> *"Implement and evaluate advanced machine learning techniques to real-world problems."*
>
> **UTS Handbook 2026, 36120, learning objective 1**

> *"In-depth knowledge of deep learning, Generative AI, and other advanced AI technologies … Experience in building and deploying AI models and solutions."*
>
> **Robert Walters ad, Data Scientist – Python, SQL, Sydney CBD, 23 Sep 2026**

Same skill, different words. Only the ad asks for generative AI, which ESCO has no concept for yet.

2 Three users, one question each

Elective chooser	Which subject moves me closest to my target role?
Job seeker	What does my resume lack for this ad?
Career switcher	How far is my profile from a new role?

Expected outcome: in under a minute, a ranked shortlist of electives that close the gap and a resume that knows its gaps.

3 Six sources, one vocabulary

3,204 Seek Australia Data Scientist ads, 2019–22	**13,960** ESCO v1.2.1 skill concepts	**8,753** O*NET 31.0 tool names ESCO lacks
70 UTS Handbook subjects, six courses	**1,355** LinkedIn US data roles, 2023–24	**2,484** Resumes: parsing test set, 50 CV–JD pairs

4 Pipeline

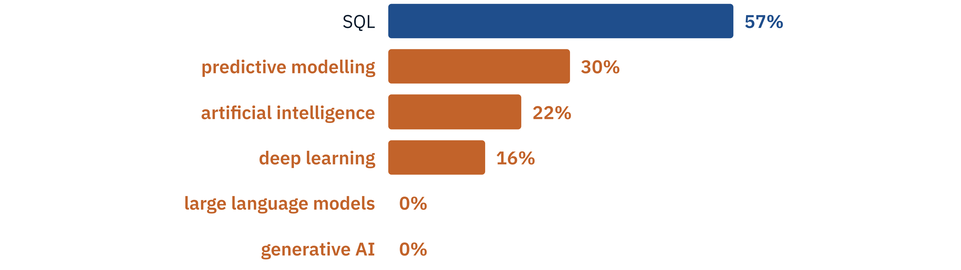
copper stages are scored in block 8

Input	Clean	Extract skills	Normalise	Match	Output
resume + ad	lowercase, strip	alias dictionary → NER, LLM	to ESCO labels	TF-IDF → embeddings	gap, subjects, chat

5 One profile against one role

8 technical skills the ad demands · **2** resume has · **6** missing

A real Sydney CBD Data Scientist ad against a synthetic early-career resume. The gaps are modelling; the two newest asks do not exist in the corpus.



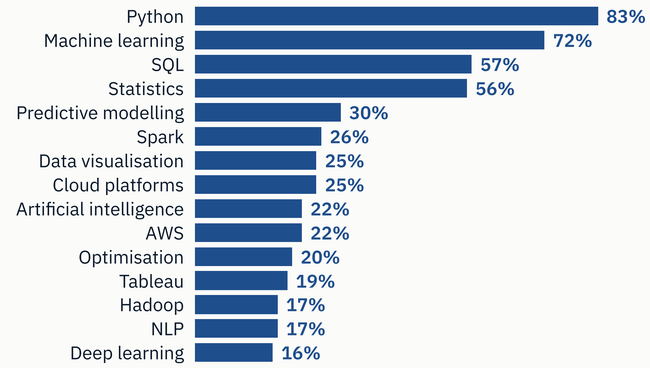
Bar = share of 3,204 Seek ads mentioning the skill or an alias, the same 138-alias dictionary as block 7. Technical skills only; the ad also asks for teamwork and problem solving. Generative AI and LLMs are 0%: the 2019–22 ads predate them.

6 Which UTS subject fills each gap

machine learning 72% **predictive modelling** 30%	**36120 Advanced Machine Learning Application** — implement and evaluate advanced machine learning techniques
artificial intelligence 22%	**36121 Artificial Intelligence Principles and Applications** — general problem-solving approaches in AI; deep neural networks
deep learning 16%	**94691 Deep Learning** — fundamentals and applications of neural networks
generative AI, LLMs 0%	No learning objective in any of the 70 subjects. 36118 Applied NLP mentions LLMs only in its description.

Keyword match over 70 UTS subjects' learning objectives, MDSI preferred; scored in block 8. Percentages are demand from block 5.

7 What Australian employers ask for



After the top four, nothing passes 31%. Off the chart, communication is in 28% of ads, more than Tableau, AWS or Hadoop.

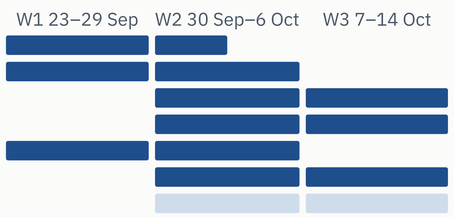
Seek Australia Data Scientist ads 2019–22, n = 3,204, 138-alias dictionary, technical skills only. LLMs: 0% here, 15.8% in 247 US Data Scientist ads 2023–24.

8 How we score each stage

Stage	Baseline → modern	Metric, test set
Extraction	ESCO dictionary → spaCy NER → GPT-4o-mini	P, R, F1; 200 hand-labelled ads
Matching	TF-IDF cosine → sentence embeddings	P@5, MRR; 50 CV–JD pairs
Subject ranking	Keyword match → sentence embeddings	P@3; 30 hand-checked skill–subject pairs
Chat	Keyword search → FAISS RAG + GPT-4o-mini	LLM judge 1–5; 30 questions

Exact ESCO matching covers only 36% of our 138-skill dictionary; 54%, including the LLM stack, have no ESCO concept.

9 Three weeks to 14 October



7 Oct pipeline end-to-end · **11 Oct** app and evaluation done · **14 Oct** Part B due

Setup · GitHub AndyPham2004/ANLP-AT2-Skillabus · Python 3.12 · spaCy, scikit-learn, sentence-transformers, FAISS · Streamlit · OpenAI API, GPT-4o-mini

Ethics · Resumes stay in memory, never on disk; names, emails and phones are masked before any LLM call. Ads used under their licences. Results are suggestions, not advice.

References · European Commission (2025), ESCO v1.2.1 · PromptCloud, Seek Australia job ads 2019–22, Kaggle · Kon (2024), LinkedIn Job Postings 2023–24, Kaggle · Bhawal (2021), Resume Dataset, Kaggle · O*NET 31.0 (2026) · UTS Handbook 2026 · Reimers & Gurevych (2019), Sentence-BERT · Lewis et al. (2020), RAG